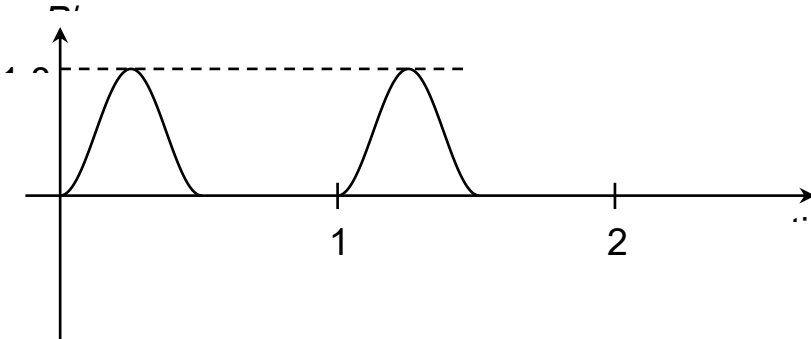


|      |                                                                                                                                                                                                                                                                                              |                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 7a   | <p>12.0 V corresponds to 3 divisions. <math>12 \div 3 = 4 \text{ V/div}</math></p> <p><math>\frac{2\pi}{T} = 314 \Rightarrow T = 20 \text{ ms}</math></p> <p>20 ms corresponds to 4 divisions. <math>20 \div 4 = 5 \text{ ms/div}</math></p> <p>(Pls do not deduct marks for d.p./ s.f.)</p> | <p>1</p> <p>1</p> <p>4</p> |
| 7bi  | <p><math>P_{pk} = \frac{V_{pk}^2}{R} = \frac{6.0^2}{20} = 1.8 \text{ W}</math></p>                                                                                                                         | 2                          |
| 7bii | <p>Horizontal line at 0.45 W.</p> <p><math>\langle P \rangle = I_{rms}^2 R</math></p> <p><math>0.45 = I_{rms}^2 (20)</math></p> <p><math>I_{rms} = 0.15 \text{ A}</math></p>                                                                                                                 |                            |

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

Sci

Equatorial plane

D1